

History, Growth and Red Flags in Paediatric Gastroenterology.

A twelve-year-old girl comes to clinic with eight months of central abdominal pain. She has no diarrhoea and no blood in her stool. She looks well. Her mother is worried, and the referral letter asks for an endoscopy. Another child, a ten-year-old boy with the same kind of pain, has been seen three times and has had a long list of blood tests, all normal. Which of these children needs more tests, and which needs a confident diagnosis and a plan? This chapter is about how to answer that question with the tools you have before any test is sent: the history, the growth chart and the examination.

The main idea is this. In paediatric gastroenterology, the history sets the probability that a child has organic disease, and the growth chart is the one objective sign that every child brings to clinic. Alarm features raise that probability, but they do not make a diagnosis. Growth, and especially the speed of growth over time, often tells you about chronic inflammation months before the gut does. And puberty is a vital sign in its own right. Expertise begins with deciding what not to test, and with noticing the slope of a growth curve that nobody has plotted.

A short note on sources. The key documents for functional symptoms are the Rome Four criteria, published in Gastroenterology in 2016. There are two paediatric papers: one for children and adolescents, by Hyams and colleagues, and one for neonates and toddlers, by Benninga and colleagues. In 2026, the Rome Foundation published Rome Five, which reorganises the childhood disorders. For growth charts, the main document is a 2010 report from the United States Centers for Disease Control and Prevention, the CDC, by Grummer-Strawn and colleagues, which recommends the growth standards of the World Health Organization, the WHO, for children under two. For inflammatory bowel disease, we use the guidelines of the European Crohn's and Colitis Organisation together with the European paediatric society, called ECCO and ESPGHAN, from 2014 and 2021. For coeliac disease we use the ESPGHAN guideline of 2020. For infant reflux we use the joint North American and European guideline of 2018, by Rosen and colleagues.

A few words first. A functional disorder, now called a disorder of gut-brain interaction, is a condition in which symptoms come from altered gut sensitivity, motility and processing in the brain, without structural or inflammatory disease to explain them. An alarm feature, also called a red flag, is a finding that raises the chance of organic disease. Sensitivity is the share of children with a disease that a test correctly picks up. Specificity is the share of children without the disease that the test correctly clears. Faecal calprotectin is a protein released by neutrophils, a type of white blood cell. When the bowel wall is inflamed, neutrophils move into it and calprotectin appears in the stool. Height velocity is the number of centimetres a child grows in a year. A z-score, also called a standard deviation score, tells you how far a measurement is

from the average for age and sex, in units of standard deviation. A z-score of minus two lies roughly at the second or third centile.

Part one. The history is the first test.

The Rome Four criteria made an important change of wording. The earlier criteria, Rome Three, required that there be no evidence of organic disease. That phrase pushed doctors to exclude everything before they could make a functional diagnosis. Rome Four replaced it with a different phrase: after appropriate medical evaluation, the symptoms cannot be attributed to another medical condition. Hyams and colleagues explain that this allows selective testing, or no testing at all, to support a positive diagnosis. It also accepts that a functional disorder can exist alongside an organic one. A child with inflammatory bowel disease in remission, or with coeliac disease on a gluten-free diet, can still have functional abdominal pain.

So the history has two jobs. It must recognise the pattern of a functional disorder, so that you can make a positive diagnosis. And it must look for the features that change the probability of organic disease, so that you know when and how to test.

Part two. The twelve alarm features.

The Rome Four paper for children and adolescents lists twelve potential alarm features in children with chronic abdominal pain. It helps to group them into four ideas.

The first idea is inherited risk. A family history of inflammatory bowel disease, coeliac disease or peptic ulcer disease raises the chance of those conditions.

The second idea is inflammation. Arthritis, perirectal disease such as fistulas or skin tags, unexplained fever and diarrhoea that wakes the child at night all point towards inflammatory disease, and towards Crohn's disease above all.

The third idea is structure and site. Persistent pain in the right upper or right lower quadrant, difficulty swallowing or pain on swallowing, and persistent vomiting suggest biliary, ileal, oesophageal or obstructive disease. Gastrointestinal blood loss belongs here too, pointing towards mucosal disease, polyps or ulcer.

The fourth idea is growth. Involuntary weight loss, deceleration of linear growth and delayed puberty all suggest chronic inflammation or malabsorption.

Notice that three of the twelve are growth items: weight, height and puberty. They are the only alarm features that you can measure rather than simply ask about. The Rome Four paper also states that faecal calprotectin appears to be better than standard tests such as C-reactive protein for separating inflammatory from functional disease.

Part three. Red flags before the age of two.

Infants and toddlers have their own set of warning signs. The Rome Four paper for neonates and toddlers defines infant regurgitation as at least two episodes a day for at least three weeks, in an infant aged three weeks to twelve months. It then lists what excludes this benign diagnosis: retching, vomiting blood, aspiration, apnoea, failure to thrive, difficulty feeding or swallowing, and abnormal posturing. Any of these means you are no longer dealing with physiological regurgitation.

The 2018 reflux guideline adds further red flags. Vomiting that starts after six months of age, or persists beyond twelve to eighteen months, suggests an organic cause. A rapidly increasing head circumference and lethargy point towards raised pressure inside the skull, not the stomach. Forceful, nocturnal, bloody or bilious vomiting, abdominal distension, rectal bleeding and chronic diarrhoea all need investigation.

Bilious vomiting deserves special emphasis. Green vomit in an infant means malrotation with volvulus until proven otherwise, and it needs a same-day upper gastrointestinal contrast study and a surgical review. In a British cohort of newborns transferred for bilious vomiting, published by Mohinuddin and colleagues in 2015, forty-six percent had a surgical diagnosis and fourteen point one percent had a time-critical surgical condition. A normal plain X-ray lowered the probability of a surgical diagnosis but did not remove it, and the authors still advised treating every case as time-critical.

Two other infant flags are worth remembering. If meconium was passed after twenty-four hours and there are other signs of obstruction, Hirschsprung disease must be excluded with a rectal suction biopsy. And if cyclic vomiting starts before the age of two, the Rome Four authors advise metabolic, neurological and anatomical testing, because a functional diagnosis is less likely in this age group.

Part four. Alarm features are not a diagnosis.

How well do alarm features actually perform? The evidence is sobering.

In 2014, Gijsbers and colleagues studied two hundred children aged four to sixteen years with recurrent abdominal pain in the Netherlands. Alarm symptoms were present in fifty-six percent of children who turned out to have organic disease, and in sixty-one percent of children with functional pain. In other words, alarm symptoms were just as common in the functional group. The authors concluded that they did not separate organic from functional abdominal pain.

In 2020, Zeevenhooven and colleagues analysed eight hundred and fifty-three children aged four to eighteen years who had had abdominal pain for more than two months. One hundred and two of them, twelve percent, had organic disease. The authors compared testing strategies. A limited set of tests, guided by alarm symptoms, picked up organic disease with a sensitivity of ninety-two percent. Adding haemoglobin and C-reactive protein to the limited panel did not

improve the ability to tell organic from functional disease.

The cost of testing everyone is real. Llanos-Chea and Saps reported in 2019 that, in one tertiary clinic, tests ordered at the first visit for abdominal pain had a diagnostic yield of only three percent.

So the right use of alarm features is to choose a targeted first test, not to order a battery. For most children with chronic abdominal pain and a reason for concern, that means coeliac serology with total immunoglobulin A, faecal calprotectin, and a stool test for *Giardia* if there is diarrhoea.

Part five. The cut-off decides what calprotectin can do.

Faecal calprotectin is the most useful single test in this setting, but its meaning depends heavily on the cut-off you choose.

In the Zeevenhooven cohort, the chart from the study showed three thresholds. Above fifty micrograms per gram of stool, calprotectin had a sensitivity of seventy-five percent and a specificity of eighty-seven percent for organic disease. Above two hundred and fifty micrograms per gram, sensitivity fell to forty-eight percent but specificity rose to ninety-nine percent. Above one thousand micrograms per gram, sensitivity was thirty-eight percent and specificity one hundred percent. So a low value is good at ruling disease out, and a very high value is close to ruling it in.

Meta-analyses in children with suspected inflammatory bowel disease give a similar picture. In 2010, van Rheenen and colleagues pooled seven paediatric studies with three hundred and seventy-one children. Sensitivity was zero point nine two and specificity zero point seven six. They calculated that if calprotectin were used first, sixty-five of every hundred children would still need endoscopy, and diagnosis would be delayed in eight percent. In 2016, Holtman and colleagues pooled nineteen studies with two thousand eight hundred and six patients and found a sensitivity of zero point nine nine and a specificity of zero point six five. They also found that symptoms alone could not rule out inflammatory bowel disease.

The practical message is that a low calprotectin is reassuring, while a value in the grey zone needs context, a repeat test, and attention to growth and to the other features, rather than an endoscopy by reflex.

Part six. Height velocity falls before the gut talks.

Now we come to growth, which is the heart of this chapter.

In 1988, Kanof and colleagues published a study of children and adolescents with Crohn's disease, looking back at their growth before diagnosis. Height velocity was already reduced before the diagnosis in eighty-eight percent. In forty-two percent, height velocity had fallen

before any intestinal symptom appeared. And in twenty-four percent, the child was growing less than four centimetres a year before the diagnosis was made.

Why is the diagnosis so often late? A British and Irish survey by Sawczenko and Sandhu, published in 2003, described seven hundred and thirty-nine new cases of inflammatory bowel disease in children under sixteen. Only about a quarter of children with Crohn's disease had the classic triad of abdominal pain, diarrhoea and weight loss, and nearly half had no diarrhoea at all. The median delay from first symptom to diagnosis was five months, and one fifth of children had symptoms for more than a year. Short stature was seen in Crohn's disease but not in ulcerative colitis.

So a child whose height velocity has slowed without an explanation should be treated as having inflammatory bowel disease until shown otherwise, even when the abdomen is quiet.

Part seven. Normal growth, year by year.

To recognise slow growth you need to know normal growth. A review by Nwosu and Lee in 2008 gives the figures. In the first year of life, a child grows about twenty-three to twenty-seven centimetres. In the second year, ten to fourteen centimetres. By the fourth year, six to seven centimetres. Before puberty, growth settles at about five to five and a half centimetres a year. During the pubertal growth spurt, girls grow about eight to twelve centimetres a year and boys about ten to fourteen.

Velocity below the twenty-fifth centile or above the seventy-fifth centile for age may be abnormal. But velocity must be measured over a long enough interval, usually six to twelve months. The reason is measurement error. A difference of a few millimetres in positioning is small against a year of growth, but large against three months of growth. A prepubertal child who is growing less than four centimetres a year is not growing normally, whatever centile the child sits on today.

Part eight. Which chart, and why it matters.

The chart you plot on is a clinical decision. There are two main kinds.

The WHO charts of 2006 are a growth standard. They describe how children should grow under optimal conditions. The children in the WHO study were all breastfed to twelve months and predominantly breastfed for at least four months, and they lived in healthy environments. The CDC charts of 2000 are a growth reference. They describe how a sample of American children did grow at one time and place, and only about half of that sample had ever been breastfed.

The CDC report of 2010 recommends the WHO charts for all children under twenty-four months of age, whatever their feeding, and the CDC charts from two to nineteen years. When

using WHO charts in the United States, the cut-offs are the two point three centile and the ninety-seven point seven centile, which correspond to two standard deviations below and above the mean. In the United Kingdom, the Royal College of Paediatrics and Child Health uses UK-WHO charts. These combine British data at birth, WHO data from two weeks to four years, and British data from four to twenty years, and they use the zero point four and ninety-nine point six centiles as their outer lines. In 2022, the CDC also published extended body mass index charts for children with severe obesity, reaching up to a body mass index of sixty.

Two practical points follow. First, correct for prematurity until at least twenty-four months of age, and many use thirty-six months when birth weight was below one thousand grams. Second, switching from one chart to another at age two can create an apparent drop in centile that is not a real change in the child.

Part nine. The chart changes the label.

The CDC report shows how much the choice of chart changes the labels we give. Between six and twenty-three months of age, low weight-for-age was found in seven to eleven percent of infants on the CDC charts but in fewer than three percent on the WHO charts. High weight-for-length under twelve months was found in nine to thirteen percent on CDC charts and five to nine percent on WHO charts, and that difference disappeared by eighteen to twenty-three months. The prevalence of short stature was similar on the two sets of curves.

The explanation is the breastfed infant. Breastfed infants grow more slowly than formula-fed infants between about three and eighteen months, and this is normal. On CDC charts, a healthy breastfed infant can look as if growth is faltering after three months. On WHO charts, which were built from exactly such infants, the same child tracks normally. So under the age of two, an underweight label on a WHO chart is more likely to mean something real.

Part ten. Crossing lines is normal in infancy.

Doctors are taught that crossing two major centile lines is a warning sign. That is true after infancy, but not before. In a study of ten thousand eight hundred and forty-four children plotted on CDC charts, published by Mei, Grummer-Strawn and colleagues in 2004, crossing two major percentile lines between birth and six months was common. It happened in sixty-two percent for weight-for-height, thirty-nine percent for weight and thirty-two percent for height. Between six and twenty-four months, it happened in thirteen to fifteen percent for height. It was least common between twenty-four and sixty months.

Infants move towards their own genetic channel during the first two years. After the second birthday, crossing centiles is no longer physiology, and it warrants evaluation.

Part eleven. The parents are the reference.

A single height must be judged against the child's genetic potential. The mid-parental height, described by Tanner and colleagues in 1970, is the simplest way to do this. For a boy, add the father's height, the mother's height and thirteen centimetres, and divide by two. For a girl, add the father's height and the mother's height, subtract thirteen centimetres, and divide by two. The target range is plus or minus eight and a half centimetres around that figure, which is roughly two standard deviations.

How should this guide referral? In 2008, Grote and colleagues developed evidence-based referral rules for short stature in the Netherlands. For children aged three to ten years, refer if the height standard deviation score is below minus two point five; or if it is below minus two point one and more than two standard deviations below the target height; or if it has fallen by more than one standard deviation and is below minus two point three. For children under three, refer if height is below minus three once, or if length is below minus two point five on two occasions within a year. In children aged three to ten, these rules picked up eighty-five point seven percent of girls with Turner syndrome and seventy-six point five percent of children with a mix of other conditions, including coeliac disease and growth hormone deficiency, while only one and a half to two percent of healthy children were referred unnecessarily. Distance from target height was the most important single criterion.

This means that a child on the twenty-fifth centile with two tall parents may be, relative to his own potential, the shortest child in the clinic.

Part twelve. Coeliac disease and short stature.

Coeliac disease can present with nothing but poor growth. A meta-analysis by Singh and colleagues, published in 2021, pooled seventeen studies with three thousand seven hundred and fifty-nine patients. Biopsy-confirmed coeliac disease was found in seven point four percent of children with short stature of any cause, about one in fourteen. In children with idiopathic short stature, meaning no other explanation had been found, it was found in eleven point six percent, about one in nine.

Family history matters too. A pooled analysis by Singh and colleagues in 2015 found coeliac disease in seven point five percent of first-degree relatives. It was eight point nine percent in siblings and three point zero percent in parents.

A Finnish study by Saari and colleagues, published in 2015, looked back at the growth of children before their coeliac diagnosis. Girls were already shorter than the reference two years before diagnosis, and boys one year before.

The ESPGHAN guideline of 2020 lists stunted growth, short stature, delayed puberty and amenorrhoea among the indications for testing with tissue transglutaminase antibodies of the immunoglobulin A class. So every short or slowly growing child should have coeliac serology,

together with total immunoglobulin A, even without a single gastrointestinal symptom.

Part thirteen. Puberty is a vital sign.

Delayed puberty is traditionally defined as no breast development by thirteen years in girls, and no testicular enlargement by fourteen years in boys. A testicular volume above three millilitres means that central puberty has begun.

Why does inflammatory bowel disease delay growth and puberty? It is tempting to blame poor eating, but that is only part of the story. In 2000, Ballinger and colleagues studied rats with colitis and compared them with healthy rats fed exactly the same amount of food. The colitic rats still grew less. About thirty to forty percent of their growth deficit was independent of undernutrition, and it worked through lower levels of insulin-like growth factor one, called IGF-1, which is made in the liver in response to growth hormone. Inflammatory molecules such as interleukin-6 lower IGF-1, and tumour necrosis factor may reduce growth hormone release.

The same group reviewed delayed puberty in inflammatory bowel disease in 2003. Puberty can be delayed even when nutrition is normal. Sex hormone levels are low while the pituitary hormones that drive the gonads are normal or high. Inflammatory mediators act directly on the axis that runs from the hypothalamus to the pituitary to the gonads. There is one consolation. Delayed bone age keeps the growth plates open longer, so if inflammation is controlled in time, catch-up growth is still possible.

How should you assess puberty? A meta-analysis by Campisi and colleagues in 2020 found that when adolescents assess their own Tanner stage using pictures, agreement with clinicians is moderate to substantial, and best at stages one and five. So a picture-based self-assessment is a reasonable screen, with examination when the answer would change management. The practical point is to ask every adolescent about puberty at every visit. A stalled Tanner stage is as informative as a falling haemoglobin.

Part fourteen. What is lost may stay lost.

Does growth recover after treatment? Not always. Sawczenko and colleagues reported in 2006 that in childhood-onset Crohn's disease, mean final adult height was two point four centimetres below the target height, and about one in five had a significantly reduced final height.

Alemzadeh and colleagues, studying one hundred and thirty-five adults with Crohn's disease in 2002, found that corticosteroid use during puberty was associated with shorter adult height. In a 2010 study by Lee and colleagues, twenty-two percent of two hundred and ninety-five children with inflammatory bowel disease had a height z-score below minus one point six four at some point, and of those with growth impairment, eighty-eight percent had Crohn's disease and twelve percent ulcerative colitis.

The guidelines respond by treating growth as a target. The ECCO and ESPGHAN guideline of 2014 lists marked growth retardation, with a height z-score below minus two point five, as a predictor of poor outcome. The 2021 update lists severe growth retardation as a reason to start anti-tumour necrosis factor therapy early, together with an immunomodulator. Steroid-sparing therapy and exclusive enteral nutrition protect the growth window.

Part fifteen. Grade malnutrition, do not just name it.

Malnutrition should be graded, because the grade sets the urgency of the plan. The consensus indicators of the Academy of Nutrition and Dietetics and the American Society for Parenteral and Enteral Nutrition, usually shortened to ASPEN, are widely used.

A weight-for-height, body mass index or mid-upper arm circumference z-score between minus one and minus one point nine is mild malnutrition. Between minus two and minus two point nine is moderate. Minus three or lower is severe. A length or height-for-age z-score of minus three or lower counts as severe. In children under two, weight gain below seventy-five percent of the expected rate is mild, below fifty percent moderate, and below twenty-five percent severe. In children aged two to twenty, weight loss of five percent of usual weight is mild, seven and a half percent moderate and ten percent severe. A fall in weight-for-length or weight-for-height z-score of one standard deviation is mild, two is moderate and three is severe. And intake of fifty-one to seventy-five percent of estimated needs is mild, twenty-six to fifty percent moderate, and twenty-five percent or less severe.

One consequence is worth stressing. A fall of one z-score is mild malnutrition even if the child still sits on the fiftieth centile. Write the grade in the note, because it sets the urgency, the dietetic plan and the threshold for supplemental feeding.

Part sixteen. What the guidelines say.

On alarm features in chronic abdominal pain, North American and European practice are aligned. Both use the Rome Four list, and the American Academy of Pediatrics and NASPGHAN technical report of 2005 points in the same direction. On a positive diagnosis with limited testing, they are aligned around the Rome Four wording. On calprotectin before endoscopy, they are aligned. Rome Four supports it for triage, and the ECCO and ESPGHAN guidelines support it for monitoring, with most of the triage data coming from Dutch studies. On growth in Crohn's disease, they are aligned, with growth as a predictor of outcome and a target of treatment. On testing short children for coeliac disease, they are aligned, and ESPGHAN makes it explicit.

The differences are in the charts. For children under two, the United States uses WHO charts with cut-offs at the two point three and ninety-seven point seven centiles, while the United Kingdom uses UK-WHO charts with outer lines at the zero point four and ninety-nine point six

centiles. For children from two to nineteen, the United States uses the CDC charts of 2000 with the extended body mass index charts of 2022, while the United Kingdom uses WHO data up to four years and its own 1990 reference after that. These are differences of convention rather than of principle.

The material change is in classification. Rome Five, published in 2026, groups the childhood disorders of gut-brain interaction by region and symptom pattern rather than by age, adds new entities such as biliary pain syndrome and centrally mediated abdominal pain syndrome, and renames infant colic as infant distress syndrome. Expect the terms in referral letters and papers to shift over the next few years.

Part seventeen. What to do differently in clinic.

Let us return to the two children from the opening.

For the ten-year-old boy with periumbilical pain, normal growth and no alarm features, the Rome Four approach supports a positive diagnosis of a functional abdominal pain disorder, with limited tests at most. More blood tests are unlikely to help. In the Zeevenhooven cohort, adding haemoglobin and C-reactive protein added nothing.

For the twelve-year-old girl, the answer depends on her growth chart. If you plot her previous heights and find that her height velocity has fallen to three centimetres a year, and her puberty has stalled, then she has two of the most important alarm features, even though her abdomen is quiet. She needs faecal calprotectin and blood tests now, coeliac serology, and a low threshold for endoscopy.

More generally, here is what to change. Plot every height and weight, and calculate height velocity over six to twelve months. Calculate the mid-parental height and judge each child against their own target. Use WHO charts under two years, and expect breastfed infants to grow more slowly between three and eighteen months. Remember that crossing centiles is common in infancy and abnormal after the second birthday. Ask every adolescent about puberty. Test every short or slowly growing child for coeliac disease. Use calprotectin with an eye on the cut-off. Treat bilious vomiting in an infant as an emergency. And grade malnutrition, so that the grade drives the plan.

Key points.

Rome Four allows a positive diagnosis of a functional disorder after appropriate medical evaluation, not after exhaustive exclusion.

Alarm features were as common in functional as in organic abdominal pain. They should guide a targeted first test, not replace a diagnosis.

Height velocity was already reduced in most children before Crohn's disease was diagnosed, often before any gut symptom.

WHO charts are the standard under two years. Crossing centiles is common in infancy but abnormal after the second birthday.

About one in fourteen short children has coeliac disease, and inflammation delays puberty directly, not only through poor intake.

Questions to test yourself.

Question one. What wording did Rome Four introduce in place of the requirement for no evidence of organic disease, and what does it allow? Answer: after appropriate medical evaluation, the symptoms cannot be attributed to another medical condition. It allows selective or no testing to support a positive diagnosis, and it accepts that a functional disorder can coexist with organic disease.

Question two. Name the three growth-related alarm features in the Rome Four list for children and adolescents. Answer: involuntary weight loss, deceleration of linear growth and delayed puberty.

Question three. In the study by Gijsbers and colleagues, how common were alarm symptoms in organic and in functional abdominal pain? Answer: fifty-six percent in organic and sixty-one percent in functional pain.

Question four. In the Zeevenhooven cohort, what were the sensitivity and specificity of faecal calprotectin above fifty and above two hundred and fifty micrograms per gram? Answer: seventy-five percent and eighty-seven percent above fifty; forty-eight percent and ninety-nine percent above two hundred and fifty.

Question five. In the study by Kanof and colleagues, in what share of children with Crohn's disease was height velocity reduced before diagnosis, and in what share did it fall before any intestinal symptom? Answer: eighty-eight percent, and forty-two percent.

Question six. Which growth charts does the CDC recommend for children under two, and why? Answer: the WHO standards, because they describe how healthy breastfed infants should grow under optimal conditions. On CDC charts, normal breastfed infants can look as if they are faltering.

Question seven. How is mid-parental height calculated for a girl, and what is the target range? Answer: add the father's and mother's heights, subtract thirteen centimetres and divide by two. The target range is plus or minus eight and a half centimetres.

Question eight. What proportion of children with short stature have biopsy-confirmed coeliac disease, and how does it change in idiopathic short stature? Answer: seven point four percent, about one in fourteen, rising to eleven point six percent, about one in nine, in idiopathic short stature.

Question nine. In the rat colitis model, what share of the growth deficit was independent of undernutrition, and through which hormone did it act? Answer: about thirty to forty percent, through lower insulin-like growth factor one.

End of chapter.